

JEE MAIN REVISION MASTER: ELECTROMAGNETIC WAVES

Subject: Physics | Target Scope: JEE Main Previous Years | Empirical Base: 166 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	EM Wave Characteristics & Spectrum	115 PYQs	EM Spectrum
Very High	Displacement Current & Maxwell's Equations	52 PYQs	Maxwell

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
EM Wave Characteristics & Spectrum	$c = \frac{1}{\sqrt{\mu_0 \epsilonpsilon_0}} = \frac{E_0}{B_0}, \quad u_{\text{avg}} = \epsilonpsilon_0 E_{\text{rms}}^2 = \frac{B_{\text{rms}}^2}{\mu_0}, \quad I = u_{\text{avg}} c = \frac{E_0 B_0}{2 \mu_0}$
Displacement Current & Maxwell's Equations	$I_d = \epsilonpsilon_0 \frac{d\Phi_E}{dt}, \quad \oint \vec{B} \cdot d\vec{l} = \mu_0 (I_c + \epsilonpsilon_0 \frac{d\Phi_E}{dt})$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 EM Wave Characteristics & Spectrum (Very High Yield — 115 PYQs)

Core Principle: Transverse nature of EM waves ($\vec{E} \perp \vec{B} \perp \vec{k}$), Poynting vector, energy density, and frequency bands of spectrum.

■■ **Caution / JEE Pitfall:** In an EM wave, the energy is distributed EQUALLY between electric and magnetic fields: $u_E = u_B$.

■ **Speed Trick:** Radiation pressure on perfect absorbing surface: $P = I/c$; on perfect reflecting surface: $P = 2I/c$.

Representative Source PYQs:

- [2026 Morning Shift] Q#1 (MCQ): "A point light source emits E.M. waves in free space. A detector, placed at a distance of L_m , measures the intensity as I_0 . The dete..." [Ans: B]
- [2026 Evening Shift] Q#1 (MCQ): "An electromagnetic wave travels in free space along the x-direction. At a particular point in space and time, $B = 2 \times 10^{-7} \text{ T}$..." [Ans: N/A]
- [2026 Morning Shift] Q#1 (MCQ): "Match the LIST-I with LIST-II } List-I List-II A. Radio-wave I. is produced by Magnetron valve B. Micro-wave II. due to change in th..." [Ans: C]

3.2 Displacement Current & Maxwell's Equations (Very High Yield — 52 PYQs)

Core Principle: Maxwell's correction to Ampere's law, displacement current inside charging capacitor dielectric space.

■ **Caution / JEE Pitfall:** Inside the charging capacitor plates, conduction current is 0 and displacement current equals I_c ; in the wire, displacement current is 0.

■ **Speed Trick:** Continuity of current holds everywhere: $I_{\text{(conduction)}} = I_{\text{(displacement)}}$.

Representative Source PYQs:

- **[2026 Evening Shift] Q#1 (Numerical):** "An electromagnetic wave of frequency 100 MHz propagates through a medium of conductivity, $\sigma=10\text{mho/m}$. The ratio of maximum conduction..." **[Ans: 1800]**
- **[2026 Morning Shift] Q#3 (MCQ):** "A displacement current of 4.0 A can be set up in the space between two parallel plates of $6\mu\text{F}$ capacitor. The rate of change of potential..." **[Ans: B]**
- **[2026 Morning Shift] Q#3 (MCQ):** "Match List - I with List - II. List - I Relation List - II Law A. $\vec{E} \cdot d\vec{l} = -\frac{d}{dt} \int \vec{B} \cdot d\vec{a}$ I...." **[Ans: B]**

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 6th April Morning Shift (Q#1)

A point light source emits E.M. waves in free space. A detector, placed at a distance of L_m , measures the intensity as I_0 . The detector is now shifted to another location on the same spherical surface ensuring the angle between original location and new location as 45° . The measured intensity at new location will be ____.

- (A) $\frac{I_0}{4}$
- (B) I_0 **[Correct]**
- (C) $\frac{I_0}{\sqrt{2}}$
- (D) $\frac{I_0}{2}$

Correct Answer: B

Solution Analysis: The intensity (I) of electromagnetic waves from a point source in free space follows the inverse square law. $I = \frac{P}{4\pi r^2}$ Where P is the power of the source and r is the distance from the source. Initial distance is L and the intensity recorded is I_0 . It is given that the detector is shifted to another location on the same spherical surface. On a spherical surface all points are at a constant distance r from the center (the source). Since the detector remains on the same spherical surface, its distance from the point source remains L . The angle (45°) represents

Benchmark Problem 2: JEE Main 2026 (Online) 21st January Evening Shift (Q#1)

An electromagnetic wave of frequency 100 MHz propagates through a medium of conductivity, $\sigma=10\text{mho/m}$. The ratio of maximum conduction current density to maximum displacement current density is ____ . $\left[\text{Take } \frac{1}{4\pi\epsilon_0} = 9 \times 10^9 \text{Nm}^2/\text{C}^2\right]$

Correct Answer: 1800

Solution Analysis: According to Ohm's Law in point form, the conduction current density is proportional to the electric field (E): $J_c = \sigma E$. The maximum value occurs when the electric field is at its peak (E_0): $(J_c)_{\text{max}} = \sigma E_0$. Displacement current density arises from a time-varying electric field. For an electromagnetic wave, the electric field can be represented as $E = E_0 \sin(\omega t)$. Maxwell defined displacement current density as: $J_d = \epsilon \frac{\partial E}{\partial t} \Rightarrow J_d = \epsilon \frac{\partial}{\partial t} (E_0 \sin(\omega t)) = \epsilon \omega E_0 \cos(\omega t)$